

Secrets Management using GPG

Preparing Secrets

A file containing API keys and passwords was created.

cat secrets.env

```
(kali㉿kali)-[~]  
$ echo "API_KEY=super-secret-key-123  
DB_PASSWORD=VeryStrongPassword" > secrets.env  
  
(kali㉿kali)-[~]  
$ cat secrets.env  
API_KEY=super-secret-key-123  
DB_PASSWORD=VeryStrongPassword  
  
(kali㉿kali)-[~]  
$
```

Encrypting Secrets

The secrets file was encrypted using GPG symmetric encryption.

gpg -c secrets.env

The original file was removed after encryption.

Decrypting Secrets

The encrypted secrets file was successfully decrypted using GPG.

gpg -d secrets.env.gpg

```
(kali㉿kali)-[~]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali㉿kali)-[~]
$ gpg -c secrets.env
gpg: WARNING: server 'gpg-agent' is older than us (2.4.8 < 2.4.9)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command "gpgconf --kill all" to restart them.

(kali㉿kali)-[~]
$ rm secrets.env

(kali㉿kali)-[~]
$ gpg -d secrets.env.gpg
gpg: AES256.CFB encrypted data
gpg: WARNING: server 'gpg-agent' is older than us (2.4.8 < 2.4.9)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command "gpgconf --kill all" to restart them.
gpg: encrypted with 1 passphrase
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali㉿kali)-[~]
$
```

Reflection

Why don't secrets belong in source code?

Secrets in source code can be leaked through repositories or accidental exposure.

Difference between symmetric and asymmetric encryption

Symmetric encryption uses one shared password, while asymmetric encryption uses a public/private key pair.

What happens if we lose the private key?

Encrypted data can no longer be decrypted.

How would this be handled in a larger enterprise?

Large companies use centralized secrets management systems, access controls, encryption, and secure key storage.